

Virtual Nature as a Digital Twin Botanically Correct 3D AR and VR Optimized Low-polygon and Photogrammetry High-polygon Plant Models

A short overview of construction methods

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ABSTRACT

Virtual nature construction methods are covered in two processes, first with low-polygon 3D plant models ideal for augmented reality (AR) and virtual reality (VR) and the second with high-polygon 3D plant models using Unreal Engine 5 and Reality Capture. Critical for scientific and information accuracy is the iterative review process with the domain expert.

CCS CONCEPTS

• Graphics Systems and Interfaces; • Visualization; • Shape Modeling;

KEYWORDS

3D Models, Botanical, Foliage, Plants, Real-time Interaction, Game Engines

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1 INTRODUCTION

Botanically correct plant models are compared to and contrasted with non-botanically correct plant models. A review of how plant inventories of individual species are combined with GIS data to create a data visualization of the landscape [Harrington et al., 2021]. The information fidelity of the virtual environment is shown to be a function of the botanical accuracy of the individual plants, the scientific information in plant and GIS data, and the expression of that information in photorealistic, high-polygon 3D models. The digital twin trustworthiness is the sum of the accuracy of information in the parts. Information fidelity and photorealistic plants, in aggregate, is shown to impact the overall trust of the final landscape.

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Two virtual nature applications, one created for AR [Harrington et al., 2019] and one created for VR from GIS data sources, field research, and botanical review will be demonstrated as examples of final immersive application projects possible. The entire production pipeline will be presented in three steps of an Expert-Learner-User Experience (ELUX) process, i.) gathering source photos in the field, ii.) artistic 3D modeling methods and techniques, iii.) iterative co-review and correction of models with the botanists and the scientific domain experts before release. Two methods of construction are presented (Figure 1). The first method uses traditional approaches to construct low-polygon 3D models. Details on how to optimize them for AR and virtual VR will be presented. Many devices with hardware constraints require low-polygon models, yet many applications desire realism. The second method uses photogrammetry techniques available in Reality Capture and Unreal Engine 5. Presented are two detailed, step-by-step, production pipelines for both methods. Source photos of plants gathered in the field will be provided for the work.

2 WHY IS BOTANICAL ACCURACY REQUIRED?

Why should you care about botanical accuracy, when everything that looks green to you, looks the same? Well that is the interesting thing about perception. It is a top down and bottom up process. You only “see” what you already know. Education is about creating those wrinkles in a student’s brain that connects some stimulus in the environment to an idea that is encoded as a word and an image in the brain. Botanical accuracy of the correct plants in the correct locations creates the character of the landscape. So, if you care about accuracy and precision, then this course might be of interest to you. That level of accuracy is important for applications that require a digital twin of natural environments, especially those used for decision support. A digital twin is a model and a simulation of reality, used for real-time data visualization, exploration and discovery, decision support, and control. It works by exploiting humans’ ability to process large amounts of data, explore vast datasets and dynamics, see patterns, and to be able to make sense of a complex world. If we really care about getting it right, then we need to get it right, even if it is about nature and something most people can’t “see” [Wandersee and Schussler, 1999]. Only a few people actually know that there are about 1.2 global plants named in the Royal Botanic Gardens, Kew, “Plants of the World Online” resource [2022], and only about 400,000 flowering species.

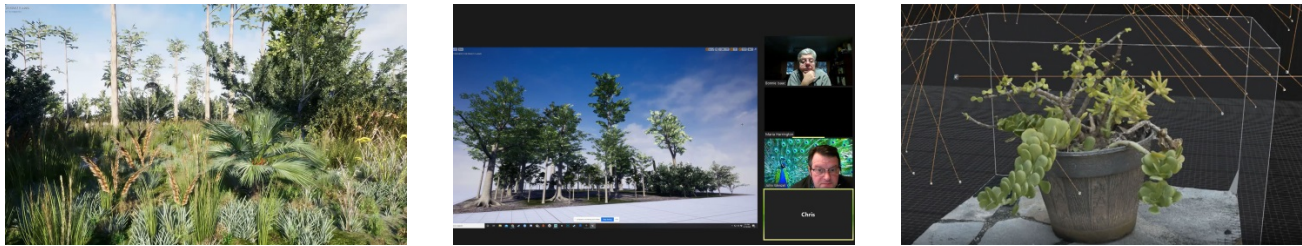


Figure 1: Left image is an example of a virtual nature. Center image is an example of the botanical review with a botanist from the Carnegie Museum. Right image is a model created with Reality Capture.

Table 1: EETA course description overview

Summary	Students will learn how to construct botanically correct 3D plant models
Learning Outcomes	Students should be able to create low-polygon and high-polygon models
Classification(s)	3D Modeling, Augmented and Virtual Reality (HCI), Visualization
Audience	Art, game design, environmental art, GIS, and computer science
Dependencies	Knowledge of photography, photo editing software, 3D modeling software, and game engines: Photoshop, Maya, Unreal Engine 5, and Reality Capture
Prerequisites	Knowledge of and past experience construction of 3D models is required. Students should already know how to use photo-editing, modeling and game engine software
Strengths	Enjoy being able to make realistic and accurate plant models
Weaknesses	Only two example plants will be covered in the pipelines, and given different biomes, the process may not translate outside of the northeastern or southeastern US
Variants	Students can use their own images or photos they gathered prior to the course
Assessment	Low-polygon models should perform well on AR and VR devices. High-polygon models should be well trimmed and connected

3 AR AND VR LOW-POLYGON MODELS

The traditional model creation process begins with gathering photographs in the field, and developing an understanding of the plant. Taking reference photos of the entire plant from multiple angles is essential for capturing the form and structure of the plant. A solid-color backdrop helps to isolate parts of the plant from the surrounding visual noise in the field. Pulling the plant components apart documents each part (stem, leaf, petal, stamen, pistil, etc.) for reassembly in the virtual model. The overall poly-count of the model is entirely dependent on plant size and complexity, but for our AR/VR optimized plants for use in Unreal 4, our largest trees ranged from 10,000 to 15,000 triangles, while smaller plants and flowers ranged anywhere from 50 to 1000 triangles.

4 PHOTOGRAMMETRY HIGH-POLYGON MODELS

Photographs taken from many different angles are used to capture enough images for the photogrammetry software to work. We recommend 500 – 1,000 for a small plant. The software will automatically stitch together the images and creates a rough 3D model. Then, some cleanup of the model will have to be done to make sure it doesn't have rough edges or gaps. These models can range from a few thousand polygons to hundreds of millions. Using Unreal Engine 5's Nanite technology, these high-polygon models present complex geometry in cinematic photorealism and dynamically and

automatically adjust gracefully to handle the Level of Detail (LOD) in real-time.

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